

Data Quality Analysis Report

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Executive Summary

This report analyzes data quality across 4 files containing 3,496 total records. The analysis identified 634 validation errors. **Key Recommendations:** • Address mandatory field completeness issues at the source • Implement data validation workflows before processing • Review critical validation errors for immediate remediation

File Quality Scores

File Name	Mandatory Complete	Total Records	Errors	Quality %
Core_Audience.xlsx	100.0%	616	11	100.0%
Core_Domain.xlsx	100.0%	21	0	100.0%
Core_Jobs.xlsx	100.0%	1,608	444	100.0%
Core_Organization.xlsx	100.0%	1,251	179	100.0%

Error Summary

Error Type	Count	Severity	Recommended Action
ENCODING_ISSUE	344	High	Source: invalid characters. Fix character encoding.
HETEROGENEOUS_TYPE	290	High	Workshop: mixed data types. Standardize approach needed.

Cross-File Referential Integrity Analysis

This section analyzes referential integrity between files using dynamic relationship discovery from Neo4j field mappings. Critical integrity issues can impact data migration success.

Organizations Domain Integrity Analysis

Business rule-based orphan detection for Organization domain relationships. This analysis identifies employee organizational assignments that lack valid organization references.

High Priority Issues (2 relationships)

Relationship	Key Field	Integrity %	Orphaned	Business Impact
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Primary Domain Name → Organizations.Organization Code	Primary Domain Name→Organization Code (via name)	22.7%	17.0	High: 17/22 employees assigned to invalid domains - impacts organizational hierarchy
Primary Organization Name → Organizations.Organization Code	Primary Organization Name→Organization Code (via name)	5.1%	561.0	High: 561/591 employees assigned to invalid organizations - impacts reporting structure

Integrity Summary

Metric	Value
Total Relationships Analyzed	2.0
Critical Integrity Issues	2.0
Total Orphaned Records	578.0

Detailed Integrity Analysis

Critical Issues (2 relationships)

Source → Target	Key Field	Integrity %	Orphaned Records	Business Impact
Orgs.Employees.Primary Domain Name → Organizations.Organization Code	Primary Domain Name→Organization Code (via name)	22.7%	17.0	High: 17/22 employees assigned to invalid domains - impacts organizational hierarchy
Orgs.Employees.Primary Organization Name → Organizations.Organization Code	Primary Organization Name →Organization Code (via name)	5.1%	561.0	High: 561/591 employees assigned to invalid organizations - impacts reporting structure

Cross-File Error Patterns (Legacy Analysis)

Error Pattern	Affected Files	Total Impact	Recommended Action
ENCODING_ISSUE	Core_Jobs.xlsx, Core_Organization.xlsx, Core_Audience.xlsx	344 records	Source: invalid characters. Fix character encoding.
HETEROGENEOUS_TYPE	Core_Jobs.xlsx, Core_Organization.xlsx	290 records	Workshop: mixed data types. Standardize approach needed.